

Sentiment Analysis on IMDB Reviews

A Project Report Submitted to Fulfill the Requirements of AI352

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Abstract

Our goal in this project is to create a model capable of analyzing, understanding, and classifying sentiments into positive and negative reviews.

In this project, we compared supervised machine learning techniques with language models. We went thru several stages, starting with processing and cleaning the text data, such as removing stop words, tokenization, lemmatization, POS tagging, and removing punctuation and numbers. And using Term Frequency-Inverse Document Frequency (TF-IDF) to extract texts and convert them into numerical representations, which reduces the impact of repeated words and gives higher weight to rare words. To prepare it for enabling the models to handle it.

We used IMDB MOVIE REVIEWS, which are user reviews and ratings in CSV format. Our results after preprocessing and cleaning the data. We chose 2 supervised learning models: logistic regression, SVM, and a large language model LLMs (Allam).

Accuracy of supervised learning models:

Logistic regression = 88.2%

SVM = 87.8%

Accuracy of large language model

LLMs model (Allam) = 95.9%

As we see, the model with the best performance is the LLMs, which indicates its ability to understand context and implicit meanings better compared to traditional models. Although traditional models still have good accuracy. These results confirmed the importance of integrating different methods to achieve a balance between accuracy and performance, and the significance of selecting the appropriate methods.

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1.Introduction

The IMDB reviews sentiment analysis dataset from the Kaggle platform was used, as sentiment analysis is one of the most important applications in NLP, aiming to extract opinions and classify them as negative or positive. This field is used in various applications such as customer review analysis and service improvement.

In this project, movie reviews were used because they contain a large number of reviews and include diverse expressions that reflect users' feelings.

The goal of our study is to use traditional machine learning methods such as logistic regression and SVM, as well as large language models that rely on context understanding and deep learning. By building models that can classify sentiments into positive and negative categories and comparing them in terms of accuracy and text comprehension ability. Our scope includes cleaning the raw data and converting it into a numerical representation using TF-IDF to reduce the impact of common words, and then training machine learning models and comparing their performance. The purpose of this is to learn linguistic patterns for sentiment classification.

2.Problem Definition

The problem lies in the need to analyze texts and build models capable of sentiment analysis and classification, as the number of texts is increasing, making manual classification difficult. The challenges for the models are that the texts are unstructured or have implicit meanings. Therefore, our project aims to build models that help classify sentiments as positive and negative.

3. Corpus Description

In this project, we use the widely recognized IMDb Movie Reviews dataset, which was obtained from Kaggle. The dataset contains a total of 50,000 movie reviews, with highly polar sentiments. It is a balanced dataset consisting of 25,000 positive reviews and 25,000 negative reviews, which helps prevent model bias during the training. Each data instance consists of a textual movie review and its corresponding sentiment label.

For our experimental implementation, the dataset was split into 80% for training the baseline models and 20% for testing their performance.

4.Related work

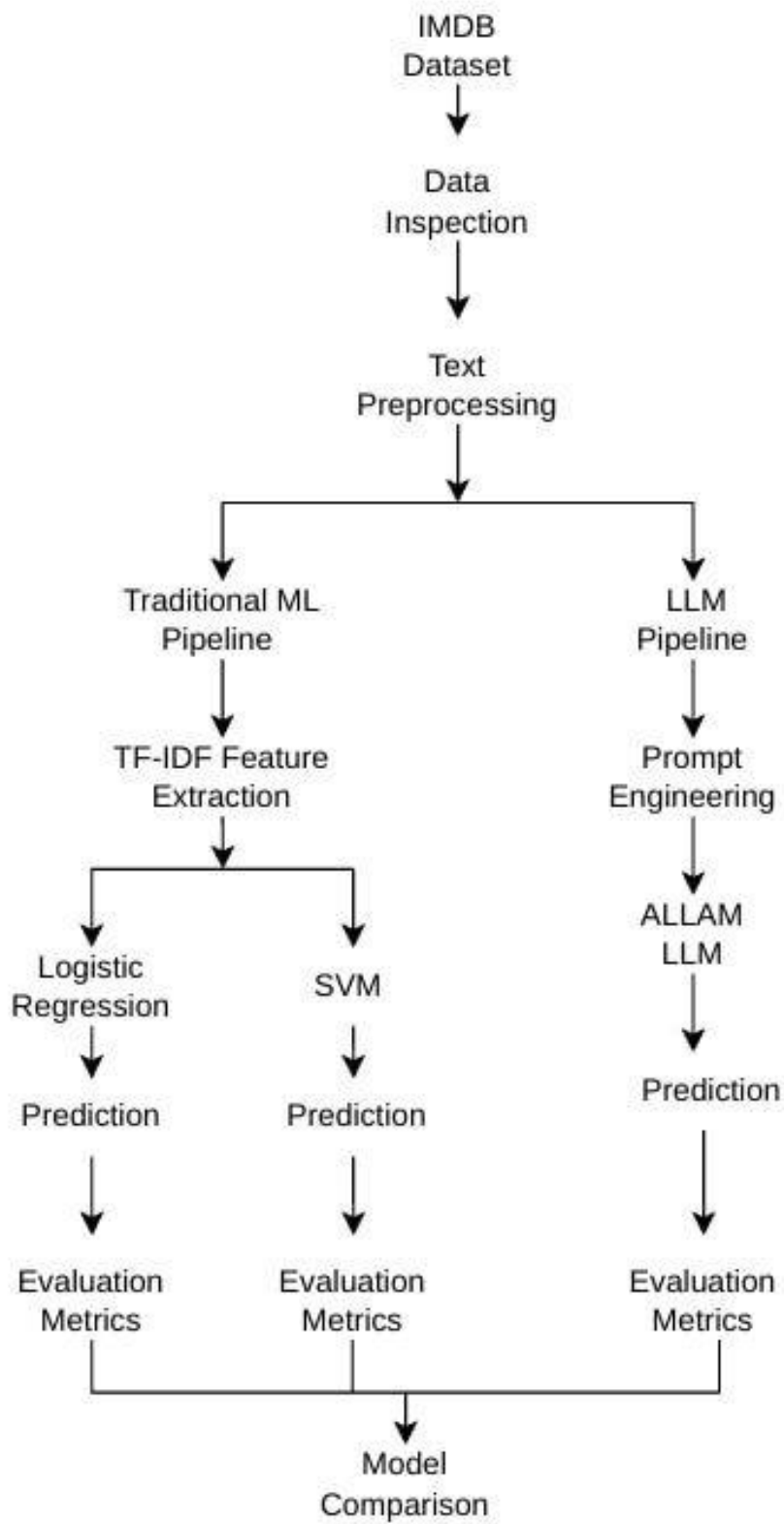
Sentiment analysis has traditionally been addressed through a wide range of approaches starting from classical machine learning approaches to advanced deep learning and large language models. Recently, a research study was conducted comparing Machine and Deep Learning models and their success in classifying sentiments within the IMDB dataset under the title **"A Comparative Analysis of Classical Machine Learning and Deep Learning Approaches for Sentiment Classification on IMDb Movie Reviews"** (Safitri et al., 2026) [3].

The results of the experiments on Machine Learning models showed that two algorithms in particular outperformed the others, with the Support Vector Machine (SVM) achieving an accuracy of 85.3% and the Logistic Regression (LR) reaching 85.1% compared to the Deep Learning models, proving that the Support Vector Machine (SVM) model combined with the Term Frequency-Inverse Document Frequency (TF-IDF) feature extraction technique remains a strong baseline.

The study's conclusion emphasized the importance of using modern Transformer-based models. Our project's contribution aligns with and builds upon this scientific recommendation, as we applied SVM and Logistic Regression algorithms as baselines, yielding comparable results that demonstrate the quality and stability of traditional models.

Subsequently, the analysis was significantly extended by integrating (ALLaM) model; a free, cost-effective, open-source, and bilingual model, which eliminates the need to bear the high financial costs associated with proprietary commercial models.

5. Pipeline Design



6. Methodology

6.1 Traditional Approach

6.6.1 Data cleaning & Text preprocessing

Preprocessing was done on the text data before training the conventional machine learning model. This is done for the purpose of improving the quality of the data and hence enhancing model performance.

Dataset was initially read using the Pandas library. There were 50,000 movie reviews in the dataset, which included 2 attributes: movie reviews and their sentiments.

The data inspection is required to ensure:

- Data type
- Number of samples
- Number of features
- Missing values
- Duplicates

There are 49,582 reviews in the dataset after removing duplicates, and there are no Missing values.

Label Encoding

The sentiment labels were converted into numerical values to make them suitable for machine learning algorithms:

```
#Encode labels
df['sentiment'].replace({'positive' : 1, 'negative' : 0}, inplace =True)
df.head(10)
```

	review	sentiment
0	One of the other reviewers has mentioned that ...	1
1	A wonderful little production. The...	1
2	I thought this was a wonderful way to spend ti...	1
3	Basically there's a family where a little boy ...	0
4	Petter Matte's "Love in the Time of Money" is...	1

Conversion to Lowercase

All reviews were converted to lowercase letters for uniformity and to minimize sparsity in the vocabulary. For instance, terms such as “Movie” and “movie” were considered the same token.

Text Cleaning and Noise Removal

The following items were eliminated using regular expressions (regex):

- HTML tags
- Punctuation
- Numbers
- Characters other than letters of the alphabet

These helped filter out noise from the text data.

Tokenization

The cleaned reviews were subjected to tokenization using NLTK tokenizer. Tokenization refers to the process by which text is divided into smaller pieces of text known as tokens (words).

Stopwords Removal

Common English stopwords such as “the”, “is”, and “at” were removed using the NLTK stopwords corpus. Removing stopwords helps the model focus on meaningful and informative words instead of high-frequency words with little semantic value.

Part-of-Speech (POS) Tagging

POS tagging was applied using the NLTK averaged perceptron tagger. Each token was assigned its grammatical category such as noun, verb, adjective, or adverb. This information was later used during the lemmatization process.

Lemmatization

Words were lemmatized using WordNetLemmatizer, which helped reduce words to their root form without losing their meaning contextually. POS tags were used to increase the efficiency of lemmatization.

Examples include:

"running" → "run"

"better" → "good"

The preprocessing phase minimized noise within the texts and enhanced feature extraction for the sentiment analysis model.

6.1.2 Feature Extraction

After preprocessing, the textual reviews were transformed into numerical vectors using the TF-IDF (Term Frequency–Inverse Document Frequency) technique.

TF-IDF is a statistical technique used for feature extraction, which calculates the significance of each term in relation to the whole set of documents. Terms that occur often in one particular review, but not in others, have high weights.

$$w = tf \times \log \left(\frac{N}{df} \right)$$

The dataset was divided into training and testing sets using an 80/20 split ratio with `random_state=42` to ensure reproducibility.

Two TF-IDF vectorizers were initialized:

- One for reviews containing stopwords.
- One for reviews after stopwords removal.

`Max_features` was set to 5000 to reduce the vocabulary size, simplify computations, and retain the most important words.

TF-IDF vectorization was applied by first fitting the model on the training dataset. Then, the `fit_transform()` method was used on the training data, while the `transform()` method was applied to the testing data.

The resulting feature matrices had the following dimensions:

Dataset	Shape
Training TF-IDF	(39665, 5000)
Testing TF-IDF	(9917, 5000)

The extracted TF-IDF vectors were then used as input features for the Logistic Regression classification model.

6.1.3 Model Development

Two traditional machine learning classifiers were implemented for sentiment classification:

- Logistic Regression (LR)
- Support Vector Machine (SVM)

Both models were trained using the TF-IDF feature vectors generated from the preprocessed movie reviews.

Logistic Regression

The Logistic Regression classifier was initialized with:

- `max_iter = 100`
- `random_state = 42`

The model was trained using both:

- reviews with stopwords
- reviews without stopwords

After training, the model predicted sentiment labels for the testing dataset.

Support Vector Machine (SVM)

The Linear Support Vector Classifier (LinearSVC) method was additionally employed for sentiment classification purposes.

The SVM model was likewise trained with TF-IDF vectors with and without stopwords removal, similarly to the Logistic Regression model.

The predictions generated by both models were evaluated using several performance metrics including:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

These metrics were used to compare the effectiveness of the traditional machine learning approaches.

6.2 LLMs approach

We use Large Language Models (LLMs) to perform sentiment classification on movie reviews. The model analyzes the semantic meaning and contextual relationships in the text to determine whether the sentiment is positive or negative

The project pretrained "humain-ai/ALLaM-7B-Instruct-preview" ALLaM model from Hugging Face Transformers was used for sentiment analysis. A prompt-based approach was applied to help the model classify whether a movie review is positive or negative.

- Converting text to lowercase
- Removing HTML tags
- Removing punctuation
- Removing extra spaces

The ALLaM model was implemented using the Hugging Face Transformers library for sentiment prediction on movie reviews. The reviews were converted into tokenized inputs before being passed to the model, and instructions were given to the model to classify each review as positive or negative during prediction.

- Using AutoModelForCausalLM and AutoTokenizer
- Evaluating performance using accuracy, classification report, and confusion matrix

7.Results

7.1 Logistic Regression Results

The Logistic Regression model achieved strong performance in sentiment classification using TF-IDF features.

The model with stopwords achieved an accuracy of **88.23%**, while the model without stopwords achieved **88.02%**.



Figure 1: Confusion Matrix of Logistic Regression (Without Stopwords)

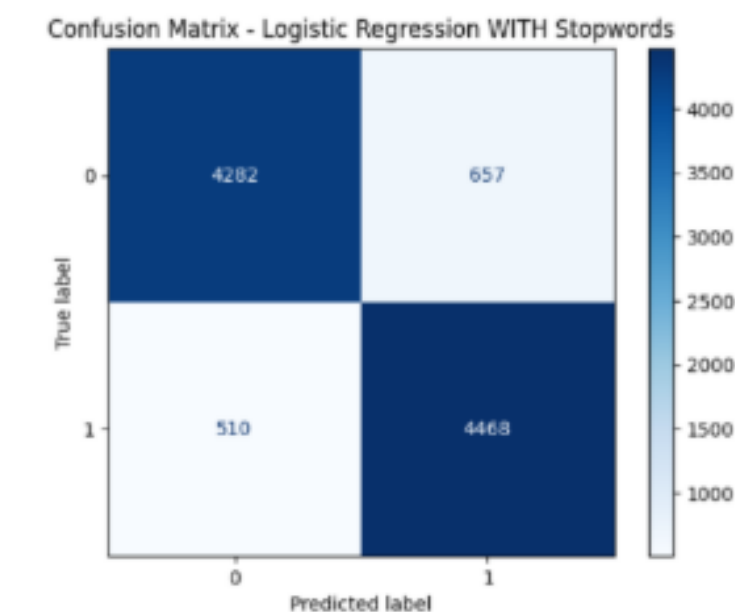


Figure 2: Confusion Matrix of Logistic Regression (With Stopwords)

The results show that keeping stopwords slightly improved the performance of the Logistic Regression model.

7.2 SVM Results

The SVM model also achieved good performance using TF-IDF features.

The model with stopwords achieved **87.82%** accuracy, while the model without stopwords achieved **87.62%**.

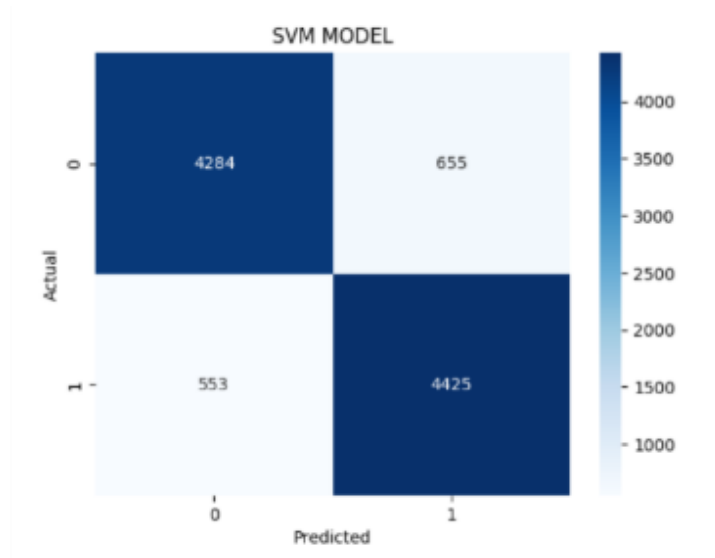


Figure 3: Confusion Matrix of SVM (With Stopwords)

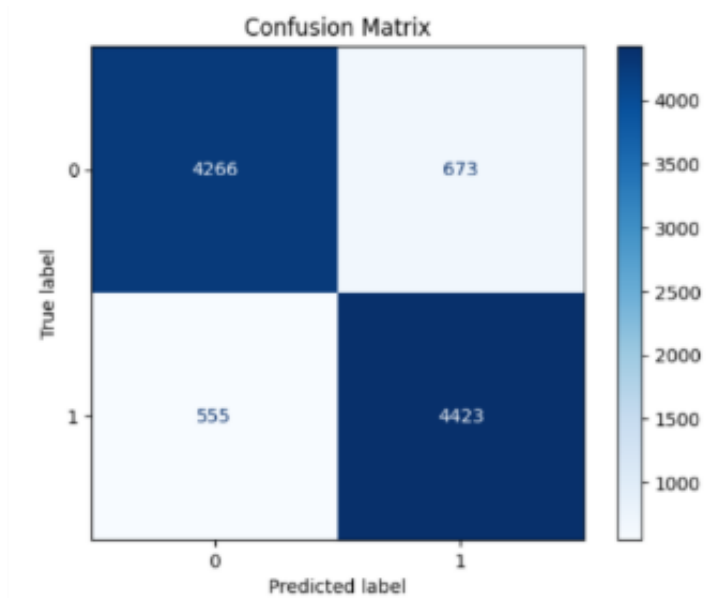


Figure 4: Confusion Matrix of SVM (Without Stopwords)

The results indicate that the SVM model also performed better when stopwords were retained.

7.3 LLMs Result

The ALLaM model achieved an accuracy of **95.9%** on the movie reviews dataset. The results showed that the model was able to classify most positive and negative reviews correctly.

The classification report showed high precision, recall, and F1-score values for both classes. The confusion matrix also showed that only a small number of reviews were classified incorrectly.

The Final, the ALLaM model showed good performance in movie review sentiment classification using the prompt-based approach.

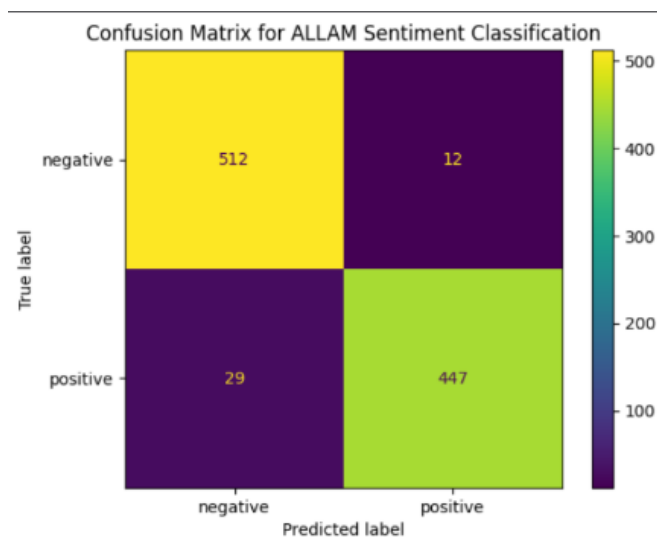
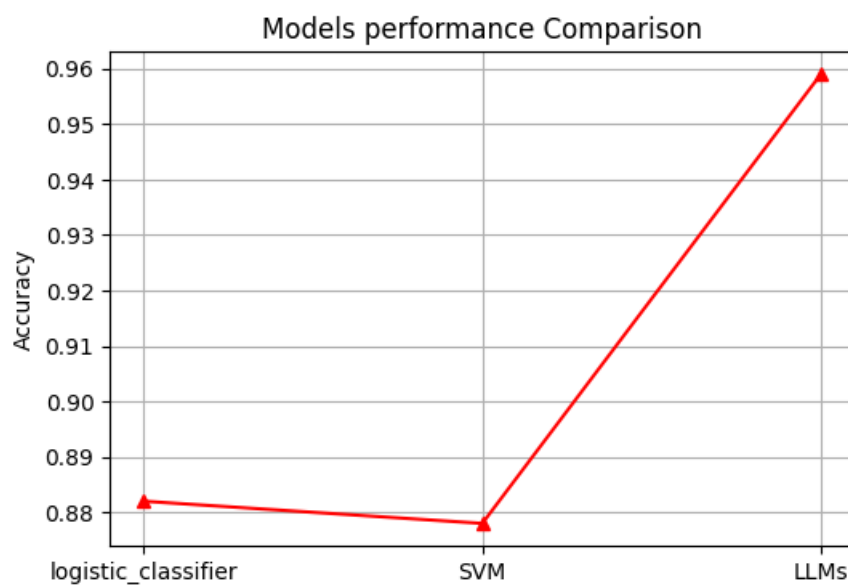


Figure 5: Confusion Matrix of the ALLaM Mode

Classification Report:				
	precision	recall	f1-score	support
negative	0.95	0.98	0.96	524
positive	0.97	0.94	0.96	476
accuracy			0.96	1000
macro avg	0.96	0.96	0.96	1000
weighted avg	0.96	0.96	0.96	1000

Figure 6: Classification Report of the ALLaM Model

7.4 Comparison Between Traditional Models and LLMs



Thru the chart, LLMs achieved the highest accuracy of 95.9%, which means they were the best at correctly classifying and were able to understand and analyze texts well.

Logistic regression ranks second with an accuracy of 88%, being a simpler model compared to LLMs, yet it still performs strongly.

SVM has the lowest accuracy of 87%, yet it is still a good and effective model.

8. Conclusion

In conclusion, this project presented a comparative benchmarking study for sentiment analysis on the IMDb movie reviews dataset, evaluating traditional machine learning models (SVM and Logistic Regression) against a modern Large Language Model (LLM) approach using ALLaM which significantly enhances contextual understanding.

Most importantly, this study proves that we do not need to rely on expensive, restrictive commercial technologies to build smart applications; accessible models like ALLaM can offer a powerful, cost-effective alternative.

For future work, this project can be expanded in several practical directions and include mixed sentiment rather than only binary (positive & negative). Additionally, Since ALLaM is a bilingual model, it would be highly valuable to evaluate its performance on Arabic or mixed English-Arabic movie reviews, which better reflect how users in our region actually write online.

9. References

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